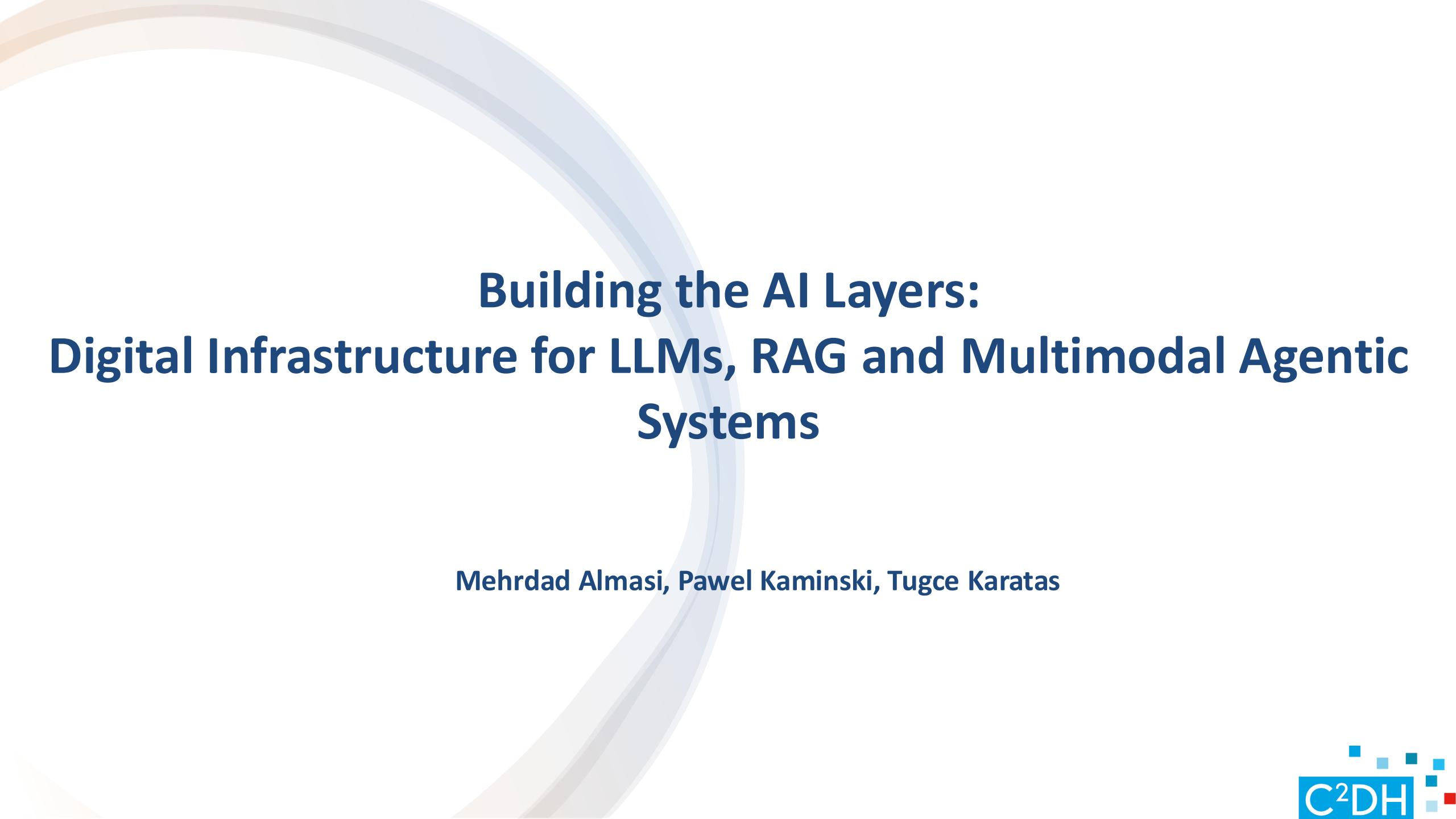




Building the AI Layers: Digital Infrastructure for LLMs, RAG and Multimodal Agentic Systems

Mehrdad Almasi, Pawel Kaminski, Tugce Karatas

Cost & Deployment Choices (nothing is truly free)

AI Service Providers



AI Service Providers

- Ready-to-use AI APIs
- No infrastructure or GPU management needed

✓ Convenient

Cloud Options



- Rent GPUs/VMs (IaaS)
- Managed inference services
- GPUs for pay-per-token/second usage

✓ Flexible

Hardware-as-a-Service



- Rent dedicated GPU servers (NaaS)
- Managed infrastructure for deploying and scaling models
- Receive bare metal access to high-performance hardware

✓ Powerful

Self-Hosting

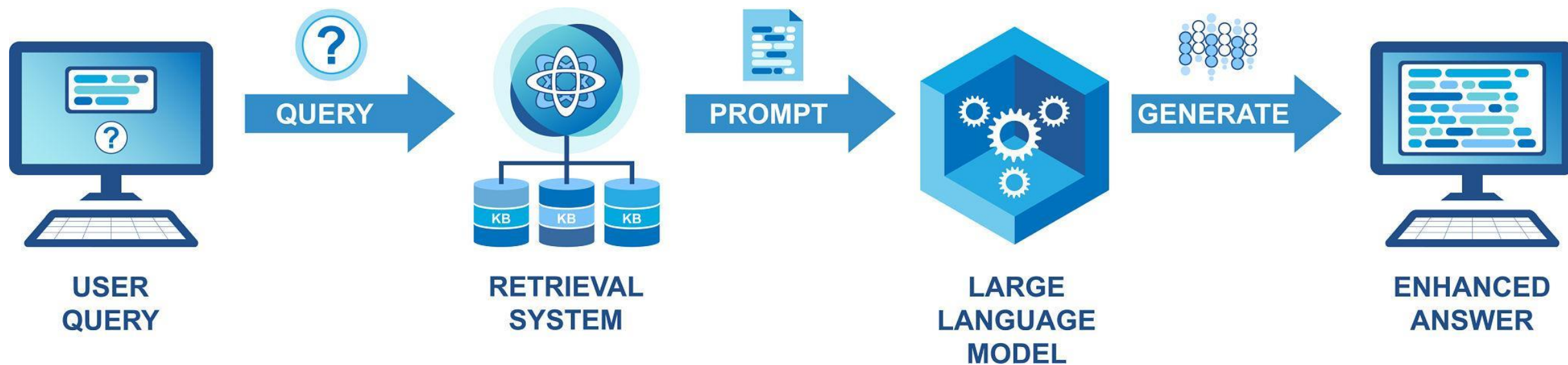


Self-Hosting

- Buy GPUs and server hardware
- Hire and manage engineers/operations teams
- Maintain, optimize, and secure your own infrastructure

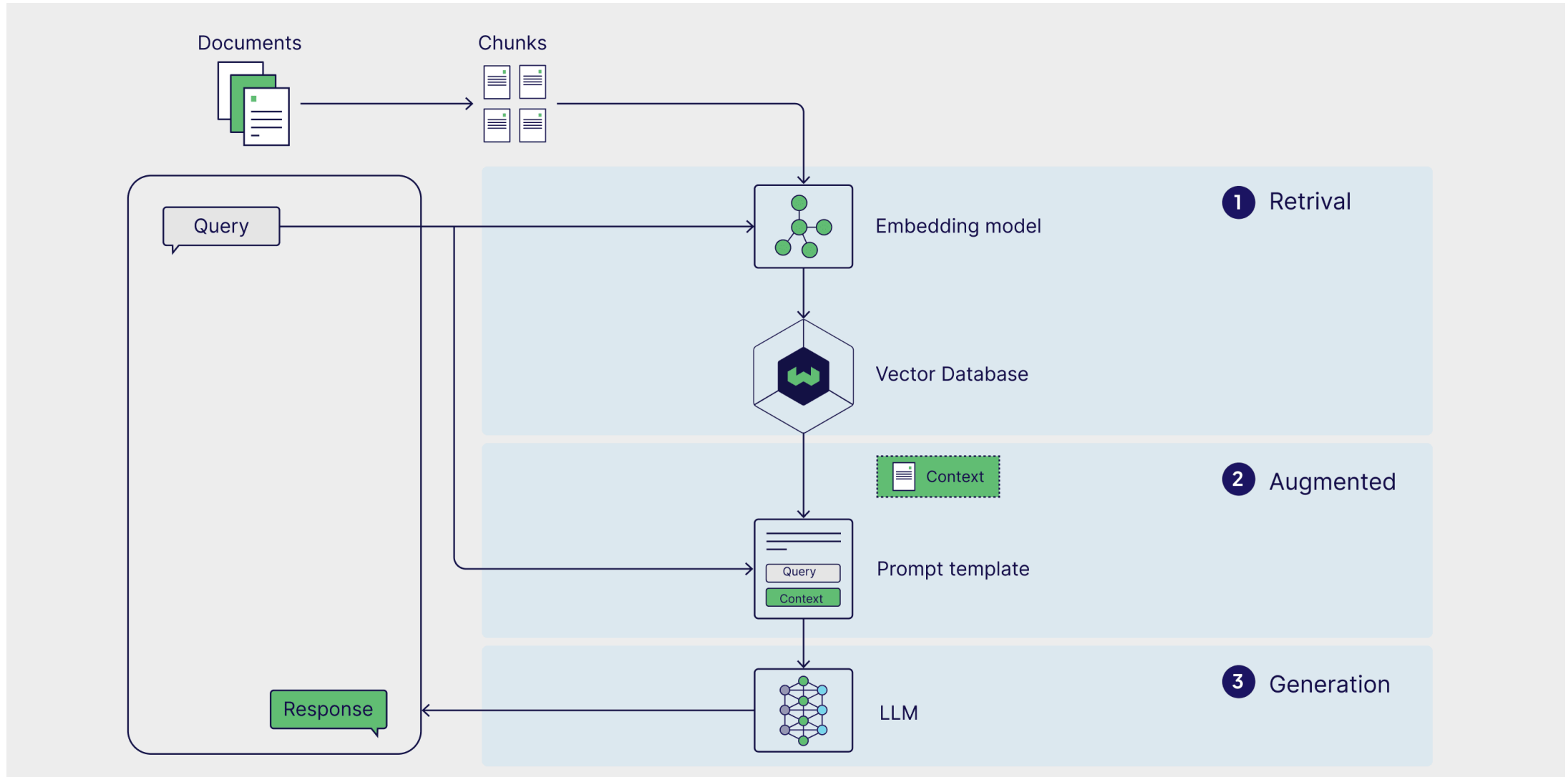
✓ Customizable

RAG



RAG (Retrieval-Augmented Generation)

- Goal: answer using **your** knowledge (docs, PDFs, web pages, databases)



Simple RAG

Retrieval-Augmented Generation

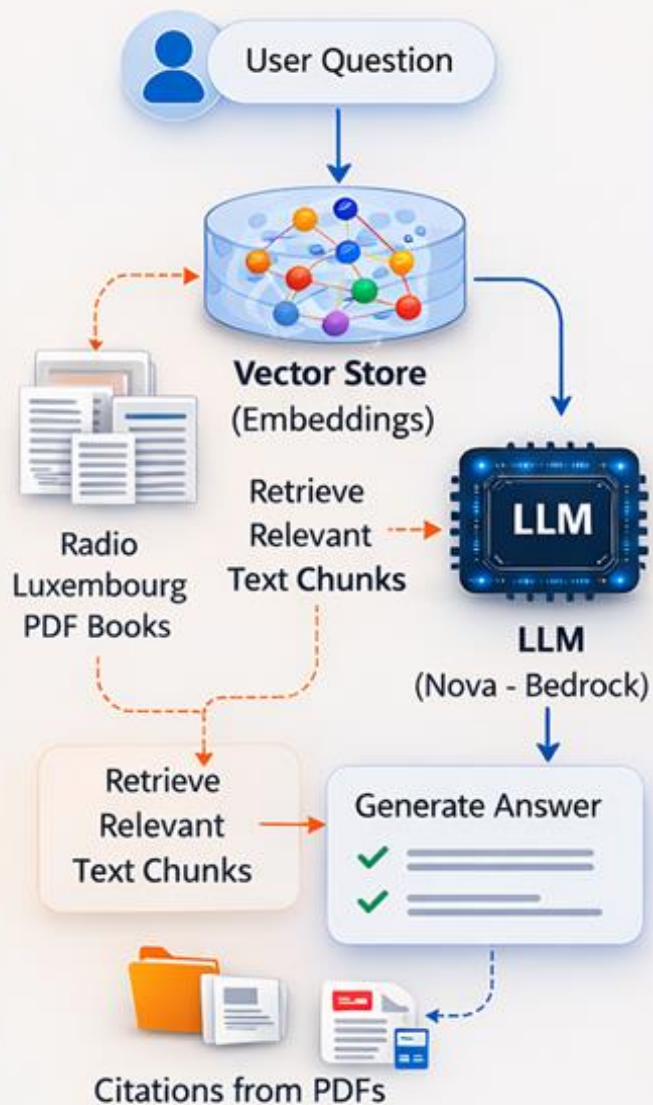
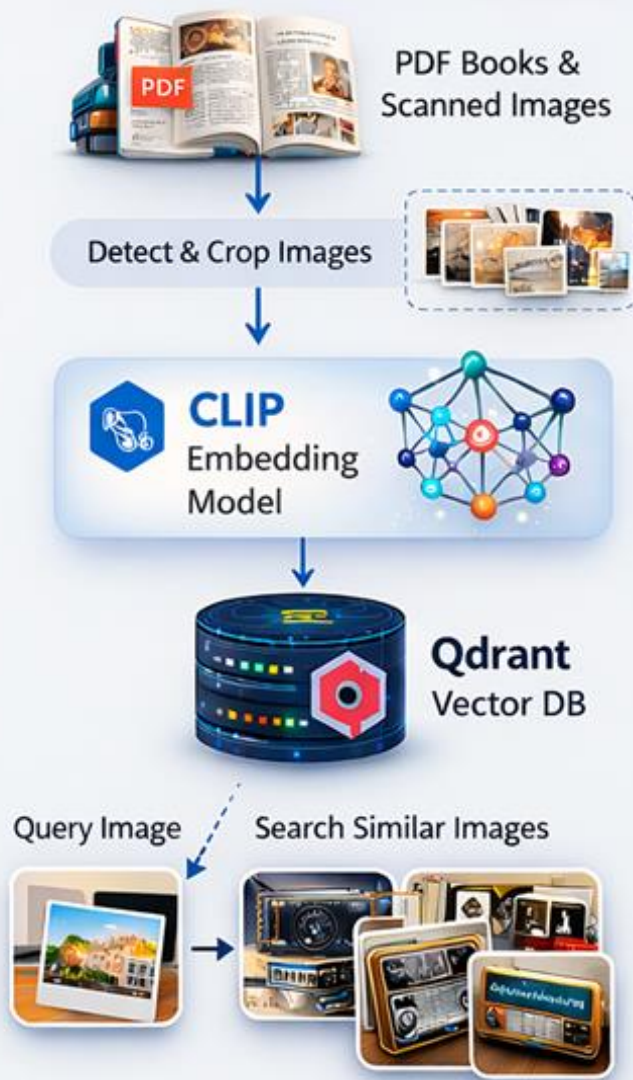


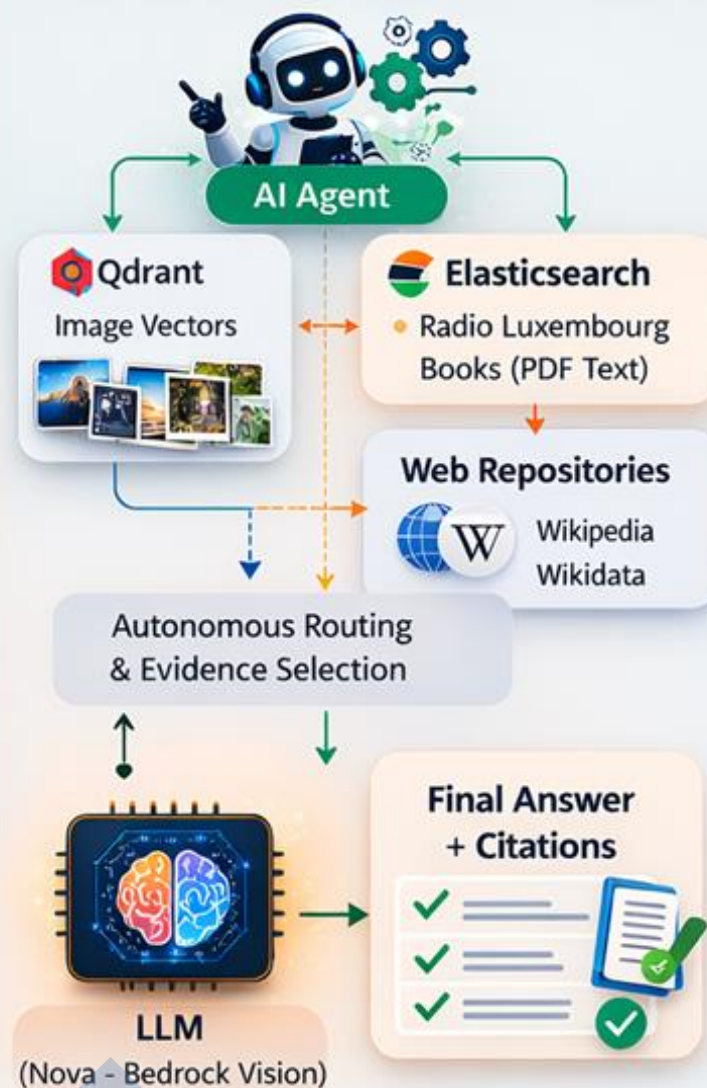
Image Retrieval System

Using **CLIP** Embeddings + **Qdrant**

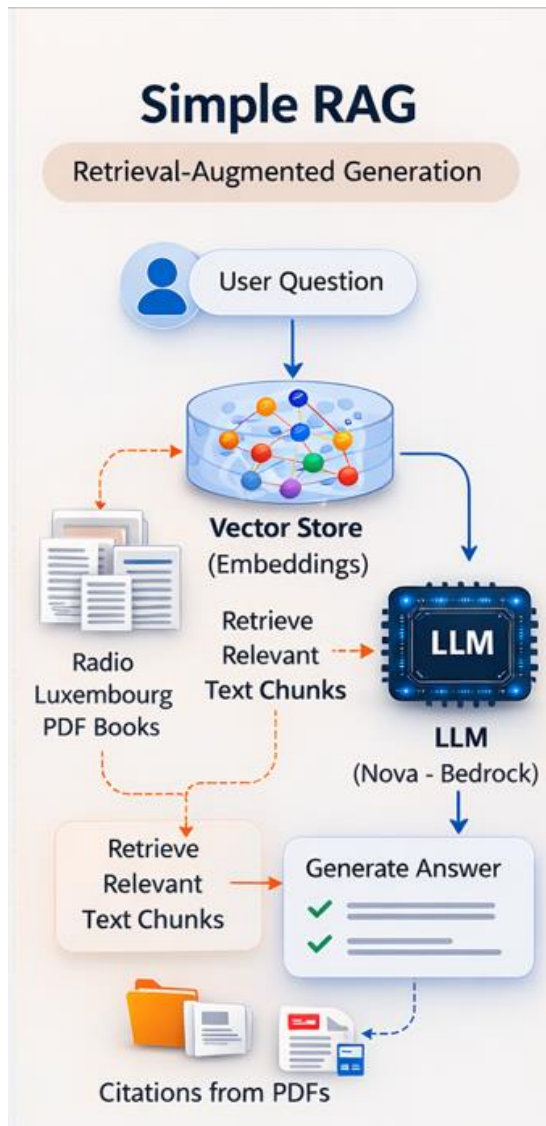


Agentic RAG

Autonomous Reasoning + Multi-Source Retrieval



Simple RAG



Radio Luxembourg semantic search

Question:

In which archives and documentation centres are Radio Luxembourg materials kept?

Search

RAG answer (validated) Citations like [1][2][5] refer to the ranked results below.

Inference output:

Radio Luxembourg materials are kept in the Archives nationales du Luxembourg, Bibliothèque centrale de Radio-France, and Archives diplomatiques du Quai d'Orsay.

- Archives nationales du Luxembourg [1][2][5]
- Bibliothèque centrale de Radio-France [1][2][3][5]
- Archives diplomatiques du Quai d'Orsay [2]

Archives nationales du Luxembourg [1][2][5]

Bibliothèque centrale de Radio-France [1][2][3][5]

Archives diplomatiques du Quai d'Orsay [2]

► Show RAG JSON

Top 5 results

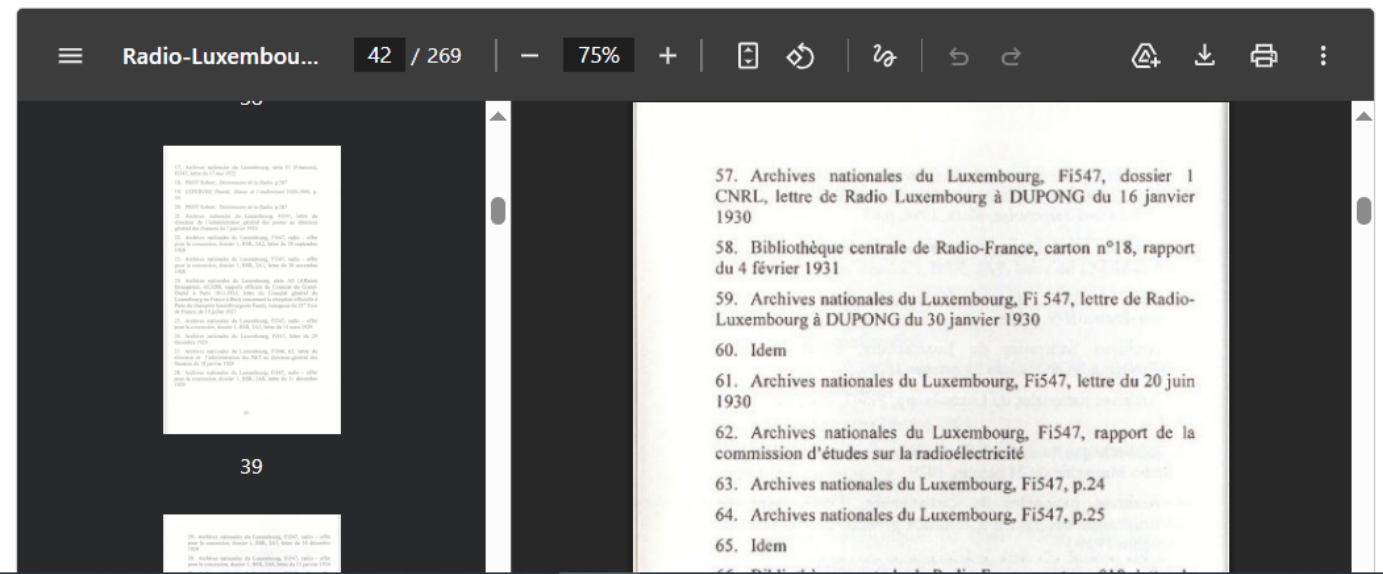
#1 (score: 1.757)

Book: Radio-Luxembourg - Histoire d'un média privé d'envergure européenne.pdf

Page: 42

57. Archives nationales du Luxembourg, Fi547, dossier 1 CNRL, lettre de Radio Luxembourg à DUPONG du 16 janvier 1930 58. Bibliothèque centrale de Radio-France, carton n°18, rapport du 4 février 1931 59. Archives nationales du Luxembourg, Fi 547, lettre de Radio- Luxembourg à DUPONG du 30 janvier 1930 60. Idem 61. Archives nationales du Luxembourg, Fi547, lettre du 20 juin 19...

[Open PDF on this page \(Jupyter\)](#)



Simple RAG

<http://13.60.254.255//search/>

Username: user2

Password: elc2dh2025esch

← → ↻ 🏠 ⚠ Not secure 13.60.254.255//search/

Radio Luxembourg semantic search

Question:

In which archives and documentation centres are Radio Luxembourg materials kept?

Search

Image Retrieval System

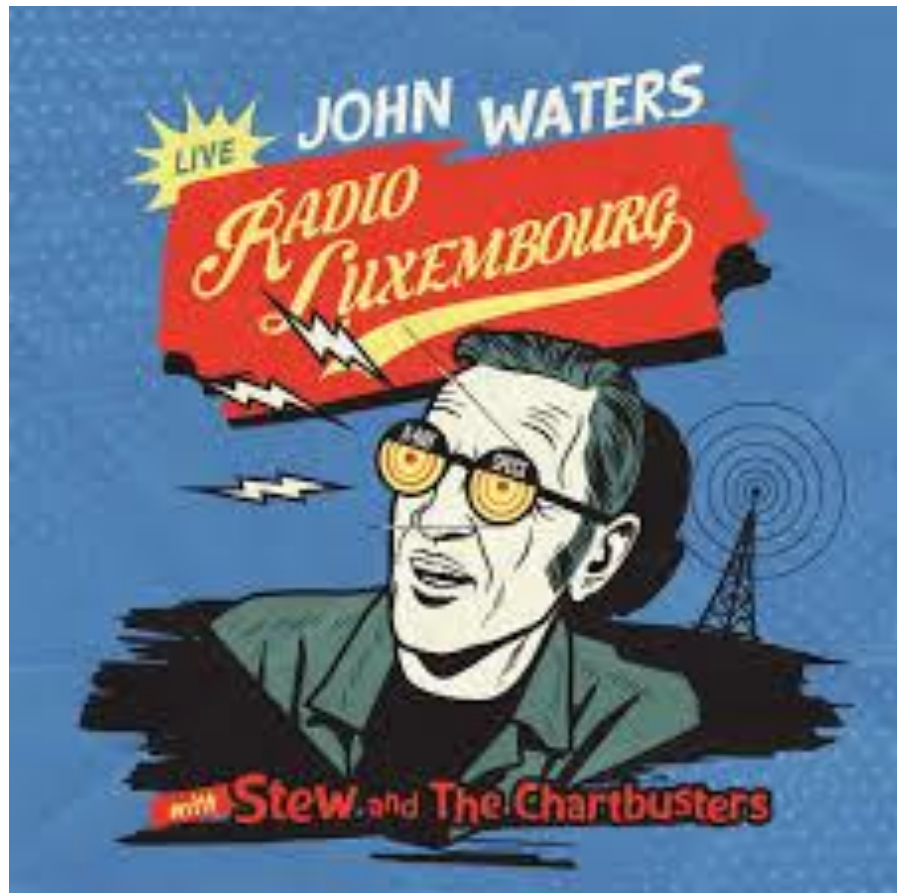


Image search results

Query Image:  Results: Showing 10 matches.

[New search](#)

Top 10 matches



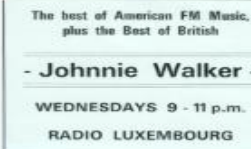
#1 (score: 0.4826)
Do: Radio Luxembourg, The station of the stars
Page: 62
File: Radio Luxembourg, The station of the stars_p0062_img01.png
[Open PDF \(page 62\)](#)
[Preview PDF \(page 62\)](#)
[Show text around this page](#)
[Debug: ES match info](#)



#2 (score: 0.4823)
Do: The Radio Luxembourg Story
Page: 1
File: The Radio Luxembourg Story_p0001_img01.png
[Open PDF \(page 1\)](#)
[Preview PDF \(page 1\)](#)
[Show text around this page](#)
[Debug: ES match info](#)



#3 (score: 0.4811)
Do: Radio Luxembourg 1979
Page: 39
File: Radio Luxembourg 1979_p0039_img02.png
[Open PDF \(page 39\)](#)
[Preview PDF \(page 39\)](#)
[Show text around this page](#)
[Debug: ES match info](#)



#4 (score: 0.4812)
Do: Radio Luxembourg 1979
Page: 87
File: Radio Luxembourg 1979_p0087_img01.png
[Open PDF \(page 87\)](#)
[Preview PDF \(page 87\)](#)
[Show text around this page](#)
[Debug: ES match info](#)



#5 (score: 0.4581)
Do: Radio Luxembourg Record Stars Book No5
Page: 69
File: Radio Luxembourg Record Stars Book No5_p0069_img01.png
[Open PDF \(page 69\)](#)
[Preview PDF \(page 69\)](#)
[Show text around this page](#)
[Debug: ES match info](#)



#6 (score: 0.4534)
Do: The Radio Luxembourg Story
Page: 156
File: The Radio Luxembourg Story_p0156_img01.png
[Open PDF \(page 156\)](#)
[Preview PDF \(page 156\)](#)
[Show text around this page](#)
[Debug: ES match info](#)



#7 (score: 0.4471)
Do: Radio Luxembourg, The station of the stars
Page: 150
File: Radio Luxembourg, The station of the stars_p0150_img01.png
[Open PDF \(page 150\)](#)
[Preview PDF \(page 150\)](#)
[Show text around this page](#)
[Debug: ES match info](#)



#8 (score: 0.4470)
Do: Radio Luxembourg Record Stars Book No5
Page: 55
File: Radio Luxembourg Record Stars Book No5_p0055_img01.png
[Open PDF \(page 55\)](#)
[Preview PDF \(page 55\)](#)
[Show text around this page](#)
[Debug: ES match info](#)



#9 (score: 0.4450)
Do: Radio Luxembourg 50 Pop Years
Page: 19
File: Radio Luxembourg 50 Pop Years_p0019_img01.png
[Open PDF \(page 19\)](#)
[Preview PDF \(page 19\)](#)
[Show text around this page](#)
[Debug: ES match info](#)



Image Retrieval System

<http://13.60.254.255/image-search/>

Username: user2

Password: elc2dh2025esch



⚠ Not secure

13.60.254.255/image-search/

Upload an image → retrieve similar images

Choose File No file chosen

Search

Results include the matched image (Qdrant+CLIP), plus prev/current/next text (Elasticsearch), and a PDF preview (served by this app).

Core Concepts Overview



Problem-Solving Agents

- **LLMs:**
 - Generates responses to user prompts
- **Agentic AI:**
 - Performs autonomous evidence retrieval, decision-making, and action execution
- **Multi-Agent Systems:**
 - Multiple AI agents collaborate to solve complex tasks



Cost & Deployment Choices (nothing is truly free)

- **AI Service Providers:**
-   **Grok**  **Claude**
by Anthropic



Convenient

Cloud Options:



Flexible

Self-Hosting:



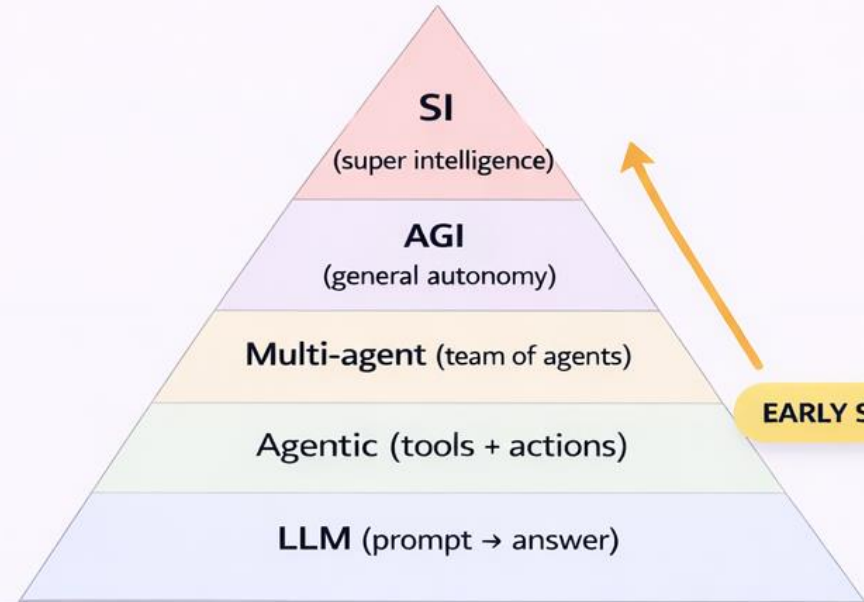
Full Control

Hardware as-a-Service



Full Control

AI Hierarchy



← Back to Search

New Search

Agentic Answer

Uploaded image:



Question:

where is this building and why is it related to Radio Luxembourg?

Answer:

Answer: The building in the image is Villa Louvigny, which served as the headquarters of Radio Luxembourg station that broadcast from Luxembourg to the United Kingdom and other parts of Europe, especially

Evidence used: repository

Confidence: high

► **Debug (router / vision / seeds)**

Top matches (repository)

Repository similarity matches — may be unrelated to the image's real identity.

score: 0.8798 **Repository item** — page 189 — [Open PDF](#)

THE RADIO LUXEMBOURG STORY 'Luxembourg lives on but bears very little resemblance to its former self.' It has only four program records right through now and assume a higher level of intelligence and age among their listeners. And 208 has its own answer to Jo Dimensions show features the latest releases on the more progressive pop scene. 127 Villa Louvigny, 2018. (Nathan Morley) One pe he banned Paul McCartney's Give Ireland Back to the Irish and McGuinness Flint's Get the People Go. At the time, Keen said there wou offend the British government. Restrictions also prevent us from presenting references, jokes or songs concerning any 127 -Belfast ...

► **Preview PDF page 189**

score: 0.8384 **Repository item** — page 248 — [Open PDF](#)

Villa Louvigny, 1961, photographie : Théo Mey, © Photothèque Ville de Luxembourg (61088-68) 248

► **Preview PDF page 248**

score: 0.8064 **Repository item** — page 246 — [Open PDF](#)

Villa Louvigny, siège de la CLR, 1937, photographie inconnu, © Photothèque Ville de Luxembourg (1937/2/328) 246

► **Preview PDF page 246**

score: 0.8055 **Repository item** — page 59 — [Open PDF](#)

THE RADIO LUXEMBOURG STORY resistance.41 Amid these incidents of resilience and courage, officials from the Nazi propaganda s addresses in the city centre. By October, they had arranged for the main German broadcasting station Reichsender to be relayed from German domestic programme relays featured a mixture of performances and music, from stars like Johannes Heesters, Herbert Ernst new Nazified radio provided both frustrations and pleasures. Most wartime programmes came from the Haus Des Rundfunks in Berlin. (Photo: Nathan Morley) A summary of the front-line reports at 11am on Sundays became essential listening. It was followed in the afternoon by the hugely popular variety show Wunschkon^ert fur die Wehrmacht...

► **Preview PDF page 59**

Wikipedia (up to 20, may be unrelated)

- [Radio Luxembourg](#) — open
- [Luxembourg](#) — open
- [Luxembourg Airport](#) — open

Wikidata (up to 20, may be unrelated)

- [Radio Luxembourg](#) — open
multilingual commercial broadcaster in Luxembourg
- [Luxembourg](#) — open
country in Western Europe
- [Luxembourg Airport](#) — open
airport in Luxembourg

Agentic RAG

<http://13.60.254.255/agentic-rag/>

Username: agenticuser

Password: elc2dh2025esch



⚠ Not secure 13.60.254.255/agentic-rag/

Agentic RAG

Upload an image + ask a question. The app will retrieve matches (Qdrant + ES) then enrich with Wikidata/Wikipedia and synthesize an answer.

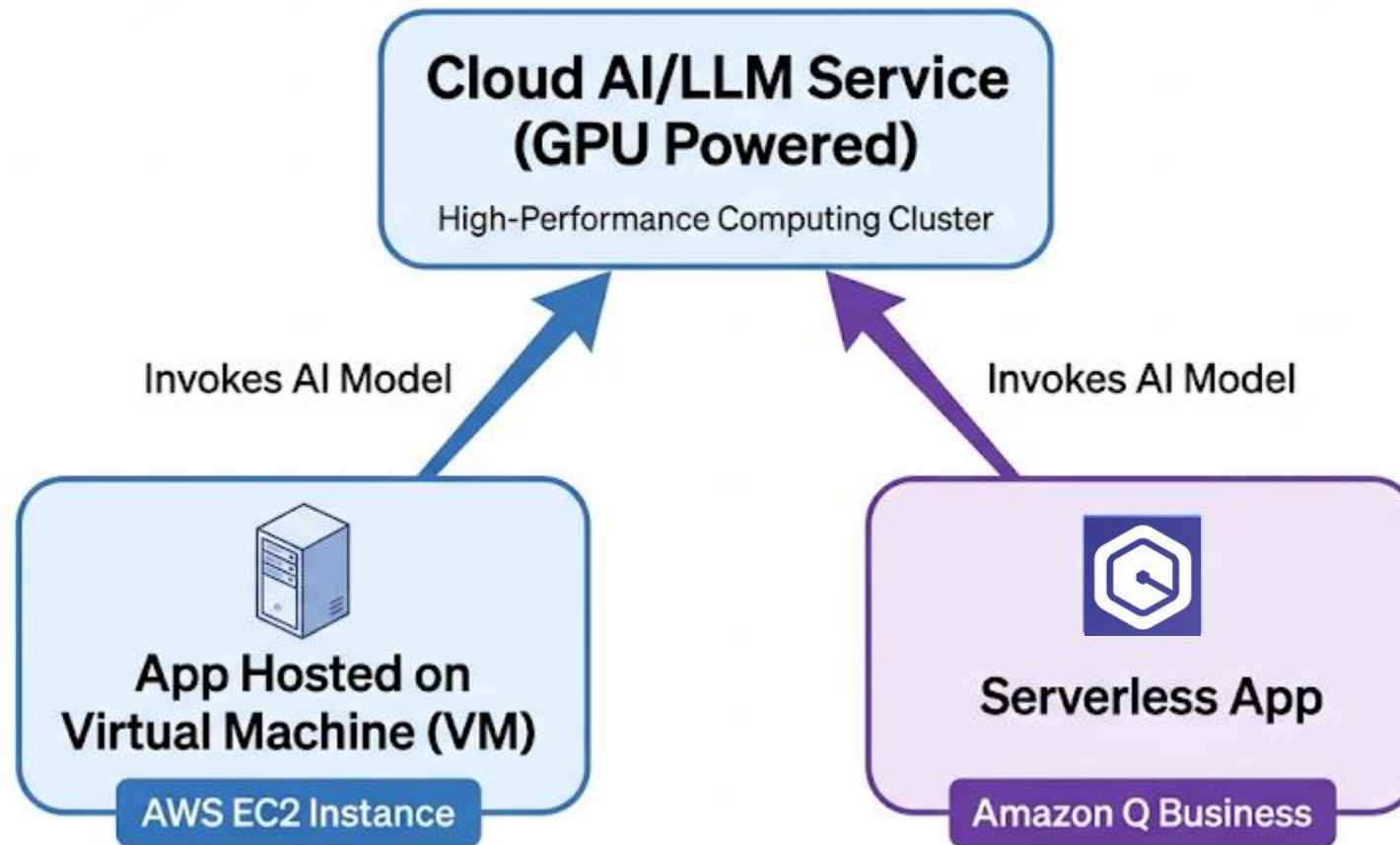
Choose File No file chosen

Example: Where is this building and why is it relevant to Radio Luxembourg?

Ask Nova

Health: /agentic-rag/health • Search-only: /agentic-rag/ /agentic-rag/search

Decreasing overhead with serverless



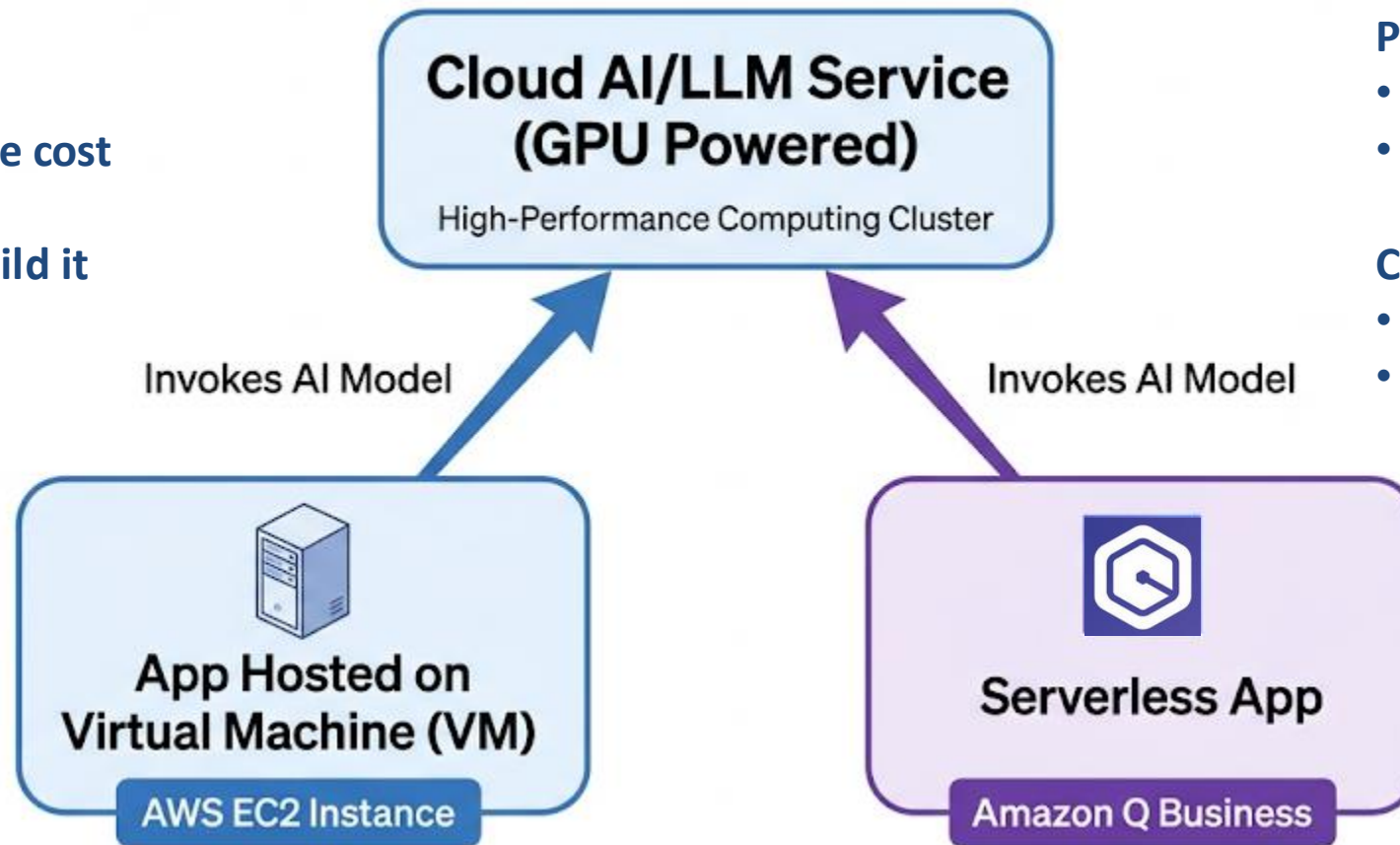
Decreasing overhead with serverless

Pros:

- Flexible
- More predictable cost

Cons:

- We'd have to build it and maintain it



Pros:

- Fast and easy to make
- 'Just works' out of the box

Cons:

- Inflexible
- Unpredictable cost

Multilingual



Quand Radio Luxembourg a-t-elle été fondée ?



Radio Luxembourg a commencé ses émissions le 15 mars 1933. [1] Cette date marque le début réel de l'existence de la station. [1] Les auditeurs à travers l'Europe ont pu entendre pour la première fois le générique familier "Bonjour, bonjour, ami fidèle, c'est Luxembourg, c'est Luxembourg, Bonjour, bonjour, c'est Luxembourg qui vous appelle !". [1]

Cependant, il est important de noter que les origines de Radio Luxembourg remontent un peu plus tôt. En 1929, une entreprise appelée Compagnie Nationale de Radiodiffusion Luxembourgeoise (émissions Radio-Luxembourg) a été créée. [1] Cette société avait pour objet la construction et l'exploitation de postes de radiodiffusion. [1]

De plus, dès 1925, un commerçant nommé François Anen avait créé une association appelée Radio-Luxembourg et mis en place un poste d'une puissance de 50 watts. [1] En 1926, la puissance de ce poste a été augmentée à 150 watts. [1]

▼ Events

▼ Sources



Wann wurde Radio Luxembourg gegründet?



Radio Luxemburg wurde offiziell am 15. März 1933 gegründet. [1] An diesem Tag begann die Station tatsächlich zu existieren und sendete für Hörer in ganz Europa. [1] Die Planungen für die Gründung einer kommerziellen Radiostation in Luxemburg begannen jedoch schon früher. Bereits 1929 wandten sich zwei französische Geschäftsleute an die luxemburgische Regierung, um die Erlaubnis für den Start eines Radiosenders zu erhalten. [2] Sie erhielten eine Lizenz, und die ersten Sendungen gingen zwei Jahre später, also 1931, auf Sendung. [2] Von Anfang an verfolgte der Sender einen internationalen Ansatz und strahlte experimentelle Programme auf Französisch, Deutsch und Luxemburgisch für einheimische Hörer und über die Grenzen in die Nachbarländer aus. [2]

Möchten Sie weitere Informationen zu einem bestimmten Aspekt der Gründungsgeschichte von Radio Luxemburg, wie zum Beispiel die Vorbereitungen vor der offiziellen Gründung oder die frühen Sendungen?



When was Radio Luxembourg founded?



Radio Luxembourg was officially founded on March 15, 1933. [1] This launch came after a group of French visionaries received a license from the Luxembourg government on December 29, 1930, to operate a commercial radio station in Luxembourg. [1]

The journey to establish Radio Luxembourg began earlier, with the station's roots tracing back to the 1920s. In 1925, François Anen, a merchant, created an association to set up a small 50-watt transmitter. [1] The following year, the power was increased to 150 watts. [1]

However, the full-fledged commercial radio station that became Radio Luxembourg came into existence in 1933. [1] This marked the beginning of its long presence in European broadcasting, despite a six-year interruption during World War II. [1]

▼ Events

▲ Sources



[1] [Radio Luxembourg 1933-1993.pdf](#)

<https://s3-datasource-slod-prod.s3.eu-west-1.amazonaws.com/audio-luxembourg/Radio%20Luxembourg%201933-1993.pdf>
Updated November 7, 2025

[2] [The station of the stars - Radio Luxembourg - 1965.pdf](#)

<https://s3-datasource-slod-prod.s3.eu-west-1.amazonaws.com/he%20stars%20-%20Radio%20Luxembourg%20-%201965.pdf>
Updated November 7, 2025

[Radio Luxembourg 1933-1993.pdf](#)

<https://s3-datasource-slod-prod.s3.eu-west-1.amazonaws.com/audio-luxembourg/Radio%20Luxembourg%201933-1993.pdf>

“ univers de l’audiovisuel, Radio Luxembourg s’apparente à un monument par le rayonnement de son audience d’abord, par sa longévité ensuite. A une de la radiodiffusion, son histoire appartient à la longue durée malgré une interruption de six années lors de la Seconde Guerre...”

Permission-based access

If a user does not have permissions to view a document, the system will refuse to retrieve information.
Here, two different users get different answers based on their access rights:



What are the strategic planning assumptions for the year 2025?



Sorry, I could not find relevant information to complete your request.

No sources



What are the strategic planning assumptions for the year 2025?



The strategic planning assumptions for the year 2025 according to your source data are:

- By 2025, 50% of data scientist activities will be automated by AI, easing the acute talent shortage. ^[1]
- In addition, I also found the following possible assumption:
By 2023, 75% of all databases will be on a cloud platform — a development that will drastically change the DBMS vendor landscape. ^[2]

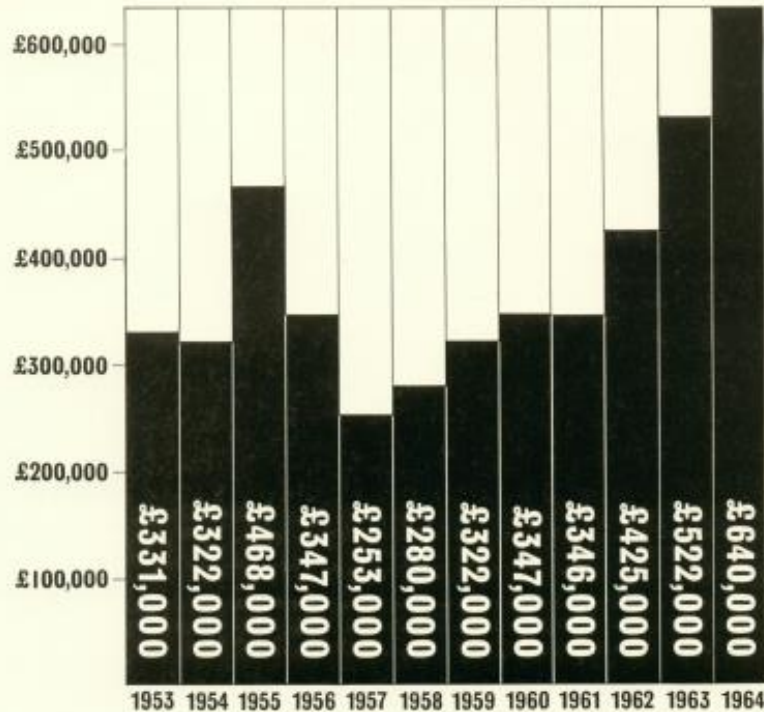
Sources ▾



^[1] [MQAIDeveloperServices.pdf](#)
 <https://demo.s3.amazonaws.com/Data/analyst-reports/MQAIDeveloperServices.pdf>

^[2] [MQOperationalDBMS.pdf](#)
 <https://demo.s3.amazonaws.com/Data/analyst-reports/MQOperationalDBMS.pdf>

AIRTIME SALES 1953 - 1964



AIRTIME RATES WERE INCREASED
IN 1964 BY APPROXIMATELY 11%

Charts are interpreted too!



What were airtime sales of Radio Luxembourg in '50 and '60?



Radio Luxembourg's airtime sales figures for the 1950s and 1960s show significant fluctuations over the years. In 1953, the sales were £331,000, which increased to £468,000 in 1955. [1] The sales then experienced a decline, reaching £253,000 in 1957. [1] As the decade progressed, there was a gradual increase in airtime sales. By 1960, the sales had risen to £347,000. [1] The upward trend continued into the early 1960s, with sales reaching £425,000 in 1962, £532,000 in 1963, and peaking at £640,000 in 1964. [1] This data demonstrates a general growth in Radio Luxembourg's airtime sales from the 1950s to the 1960s, with the latter part of the 1960s showing particularly strong performance.

▼ Events

▲ Sources



[1] [📄] [The station of the stars - Radio Luxembourg - 1965.pdf](#)

<https://s3-datasource-slod-prod.s3.eu-west-1.amazonaws.com/he%20stars%20-%20Radio%20Luxembourg%20-%201965.pdf>

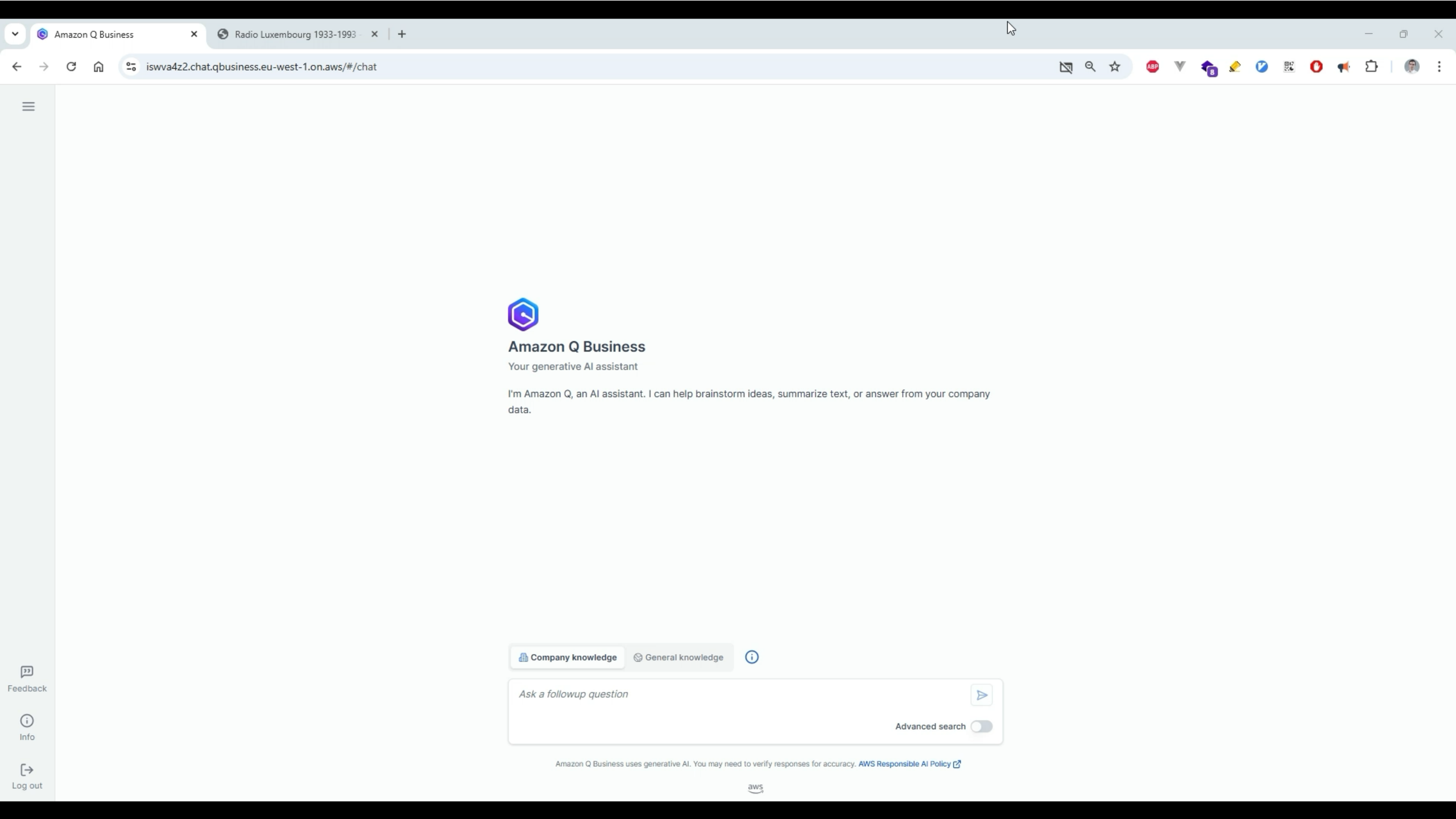
Updated November 7, 2025



THE RADIO LUXEMBOURG AUDIENCE

Radio listening, both in the United States and in this country, has made a remarkable recovery during the past ten years. Not so many years ago radio stations in the States were going for a song and then they started being

Example: Radio Luxembourg - The station of the stars. 1965.



Amazon Q Business

Your generative AI assistant

I'm Amazon Q, an AI assistant. I can help brainstorm ideas, summarize text, or answer from your company data.

Company knowledge

General knowledge



Ask a followup question



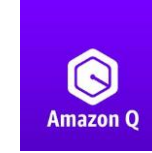
Advanced search ☐

Amazon Q Business uses generative AI. You may need to verify responses for accuracy. [AWS Responsible AI Policy](#)

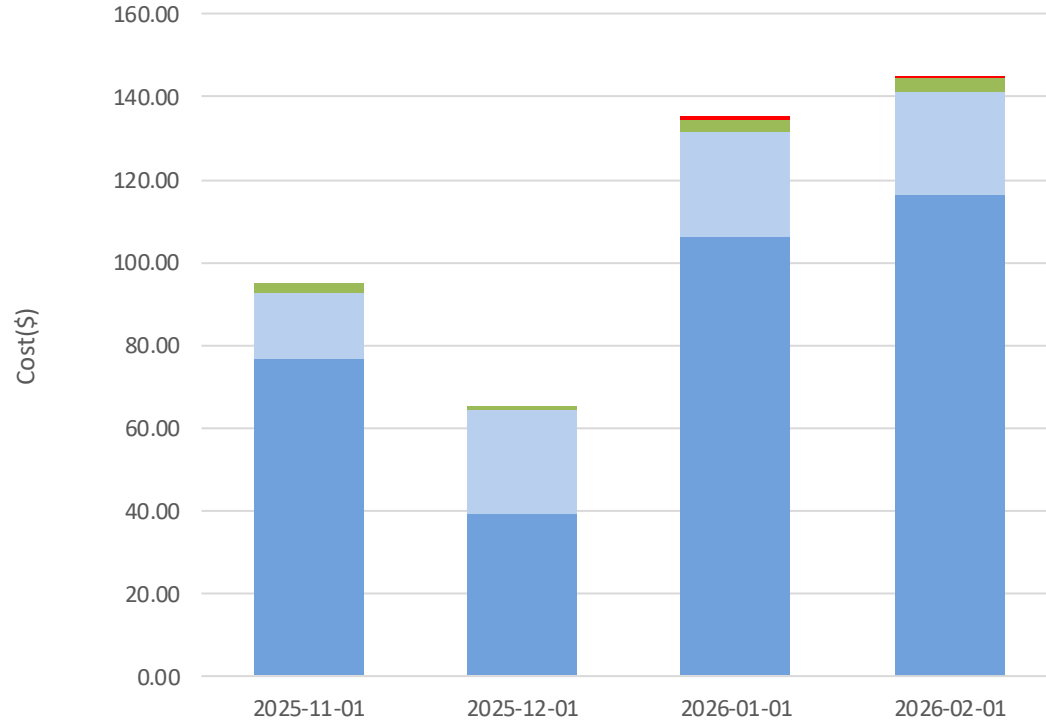




Cost

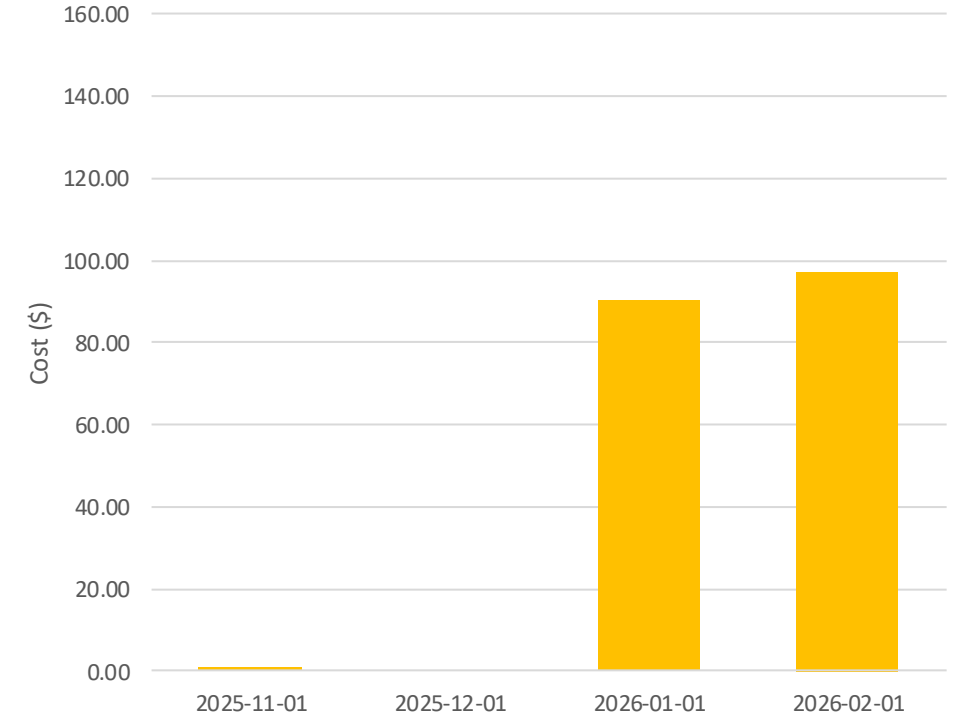


Cost of RAG on EC2 (AWS virtual machine)



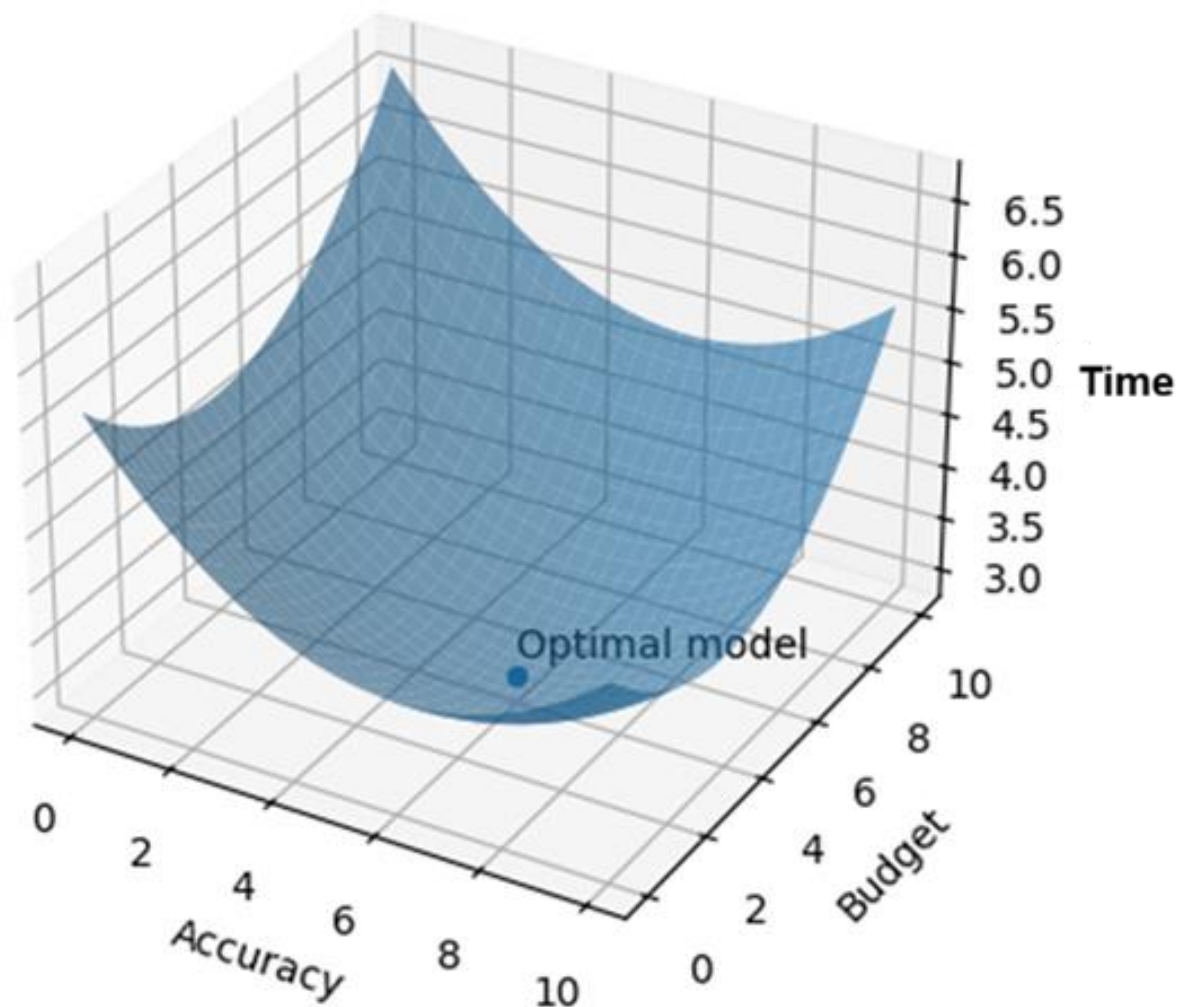
	2025-11-01	2025-12-01	2026-01-01	2026-02-01
Bedrock(\$)			0.57	0.01
VPC(\$)	2.23	1.13	3.07	3.36
EC2-Other(\$)	15.54	25.08	25.38	25.08
EC2-Instances(\$)	76.99	39.06	106.12	116.12

Cost of RAG on AWS serverless service



	2025-11-01	2025-12-01	2026-01-01	2026-02-01
Q(\$)	0.61	0.00	90.22	97.08

Choosing a model: budget × time × accuracy



Practical rule of thumb

- Higher accuracy usually costs more (API or bigger GPUs)
- Bigger context (more input text) increases latency + cost
- Commercial APIs can be fastest to start
- Self-hosting can be cheaper at scale but needs operations effort

THANKS

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